# TRACK INSPECTION REPORT

Section: Bengaluru City (SBC) - Whitefield (WFD) | Double Line, BG 1676 mm

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| **Document No.** | SWR/ENGG/TIR/2026/0412 |
| **Inspection Date** | 14 October 2026 |
| **Inspecting Official** | Sr. DEN/Co-ordination (fictional) |
| **Section Length** | Km 0.000 to Km 21.350 (UP and DN lines) |
| **Max. Sectional Speed** | 110 km/h (Goods 75 km/h) |
| **Traffic Density** | 24.6 GMT/annum |

## 1. Scope and Purpose

This report records the findings of the periodic track inspection carried out as per the Indian Railways Schedule of Dimensions and the Indian Railways Permanent Way Manual (IRPWM). The inspection covered rails, sleepers, fastenings, ballast profile, drainage, curves, turnouts, and level crossings.

## 2. Track Structure Summary

| Parameter | Specification | Observed | Status |
|---|---|---|---|
| Rail section | 60 kg/m, 90 UTS | 60 kg/m, 90 UTS | Compliant |
| Sleepers | PSC, 1660/km (M+7) | 1660/km | Compliant |
| Fastening | Elastic Rail Clip (ERC) Mk-III | ERC Mk-III, 3 missing at Km 7.2 | Action required |
| Ballast depth | 350 mm below sleeper | 320-360 mm | Deficient at 2 locations |
| Gauge (nominal) | 1676 mm (+6 / -3 mm) | 1673 to 1680 mm | Within tolerance except Km 12.4 |
| Welding | SWR, LWR, 130 m panels | LWR throughout | Compliant |

## 3. Geometry Measurements (Track Recording Car Data)

| Km | Gauge (mm) | Cross-level (mm) | Twist (mm/3.6m) | Alignment (mm) | Remark |
|---|---|---|---|---|---|
| 2.150 | 1676 | +2 | 3 | 2 | OK |
| 5.600 | 1675 | -3 | 4 | 3 | OK |
| 7.200 | 1678 | +4 | 6 | 4 | Monitor |
| 9.850 | 1677 | -2 | 5 | 3 | OK |
| 12.400 | 1681 | +7 | 9 | 6 | Gauge exceeds limit - attend within 7 days |
| 15.300 | 1674 | +1 | 3 | 2 | OK |
| 18.750 | 1676 | -3 | 4 | 5 | OK |

## 4. Defects and Recommended Actions

| Ref | Location | Defect | Priority | Action / Target |
|---|---|---|---|---|
| D-01 | Km 12.400 UP | Gauge widening to 1681 mm | High | Re-gauge, replace worn liners - 7 days |
| D-02 | Km 7.200 DN | 3 ERC missing, 2 loose | Medium | Replenish and tighten - 15 days |
| D-03 | Km 3.050 - 3.200 | Ballast starved shoulder | Medium | Ballast supplement 180 cum - 30 days |
| D-04 | Km 16.800 LC-41 | Rubber pad wear at level crossing | Low | Replace during next block - 45 days |

## 5. Curve Register Extract

| Curve No. | Km Range | Direction | Radius (m) | Degree | Cant (mm) | Cant Deficiency (mm) | Speed (km/h) |
|---|---|---|---|---|---|---|---|
| C-3 | 4.100-4.620 | LH | 900 | 1.94 | 65 | 55 | 85 |
| C-7 | 8.300-8.900 | RH | 1200 | 1.45 | 50 | 60 | 95 |
| C-11 | 13.200-13.750 | LH | 650 | 2.68 | 80 | 75 | 75 |
| C-14 | 17.050-17.300 | RH | 1500 | 1.16 | 40 | 45 | 110 |

Formula used for equilibrium speed: $V_{max} = 0.27 \times \sqrt{((C_a + C_d) \times R)}$, where $C_a$ is applied cant and $C_d$ is permissible cant deficiency, both in mm, and R is radius in metres.

## 6. Level Crossing Status

| LC No. | Km | Class | Interlocked | TVU (x1000) | Remark |
|---|---|---|---|---|---|
| LC-38 | 14.220 | Special | Yes | 42.1 | Boom and lights functional |
| LC-41 | 16.800 | A | Yes | 28.7 | Pad replacement due |
| LC-44 | 19.500 | B | No | 11.4 | Proposed for ROB |

Remarks: Overall track condition is satisfactory. Priority defects D-01 and D-02 to be attended and compliance reported to the Divisional office.

Signed: _____ Date: 14-10-2026